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Name:

Without deadlock

```
≡ text.txt
1 1 0 1 CPU{5} IO{8} CPU{5}
2 2 3 5 CPU{50} IO{3} CPU{R[2],6,F[2]}
3 3 7 3 CPU{8} IO{4} CPU{R[1],5,F[1]}
4 4 7 3 CPU{R[2],9,F[2]} IO{2} CPU{R[1],3,F[1]}
5 6 32 2 CPU{R[1],9,F[1]}
6 5 12 2 CPU{10,R[1],5,R[2],8,F[2],3,F[1],4}
```

```
> & C:/Users/sleet/AppData/Local/Programs/Python/Python313/python.exe c:/Users/sleet/Desktop/ibra/Ibraheem
[PID,wait time,IO time,Ready_time] [[1, 0, 8, 42], [2, 0, 3, 38], [3, 0, 4, 67], [4, 0, 2, 73], [6, 0, 0, 28]]
GanttChart[PID,CPU_time] [[1, 5], [2, 50], [1, 5], [6, 9], [3, 5], [4, 5], [3, 3], [4, 4], [3, 5], [4, 3], [2, 6]]
avg_waiting_time = 49.6
avg_turnaround_time = 73.0
PS C:\Users\sleet\Desktop\ibra>
```

With deadlock

```
≡ text.txt
1 1 0 1 CPU{R[1],30,R[2],15,F[1],F[2]}
2 2 0 1 CPU{R[2],15,R[3],20,F[2],F[3]}
3 3 0 1 CPU{R[3],5,R[1],15,F[3],F[1]}
```

```
[PID,wait time,IO time,Ready_time] [[1, 34, 0, 20], [2, 29, 0, 20], [3, 20, 0, 24]]
GanttChart[PID,CPU_time] [[1, 5], [2, 5], [3, 5], [1, 5], [2, 5], [1, 5], [2, 5], [1, 15], [3, 15], [2, 20], [1, 15]]
avg_waiting_time = 21.333333333333332
avg_turnaround_time = 82.33333333333333
```